

[JPJC/2024/Q4]

4 Name your **Jupyter Notebook** as:

TASK4_<your class>_<your name>.ipynb

The Airport Flight Information System plans to develop a web application to allow visitors to view information of flights arriving at or departing from Singapore Changi Airport. The database will have one table to store this information.

The fields in the `Flight` table are as follows

`Flight`:

- `flightNum`: a unique string assigned to the flight
- `departure`: the location of the airport the plane departs from
- `destination`: the location of the airport the plane arrives at
- `departureTime`: the time the plane departs from the airport
- `arrivalTime`: the time the plane arrives at the airport

Task 4.1

Write a Python program that uses SQL code to create the database `Airport` with the table given. Define the primary key for the table.

[3]

Task 4.2

The text files `TASK4.txt` store the comma-separated values for the table in the database.

Write a Python program to read in the data from the file and store in the `Flight` table in the database.

[5]

Run your program to test the data has been entered into the database correctly.

[1]

Save your Jupyter Notebook for Task 4.

Task 4.3

Write a Python program and the necessary files to create a web application. The web application has the following menu options.

View all Arrivals

View all Departures

Query Flight

Save your Python program as:

TASK_4_3_<your class>_<your name>.py

with any additional files/ subfolders in a folder named:

TASK_4_3_<your class>_<your name>

[3]

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Task 4.4

Write a Python program and the necessary files to create a web application that will display the information for all flight arrivals in Singapore.

The flights should be displayed on the web page labelled **Arrival**. It is arranged in chronological order of the arrival time, in a table with the following table header:

- Time (that is, arrival time)
- From (that is, departure location)
- Flight (that is, flight number)

The resulting web page should be accessed from the “View all Arrivals” menu option from **Task 4.3**.

Save your Python program as:

TASK_4_4_<your class>_<your name>.py

with any additional files/ subfolders in a folder named:

TASK_4_4_<your class>_<your name>

[6]

Run the web application. Save the arrival web page output as:

TASK_4_4_<your class>_<your name>.html

[1]

Task 4.5

Write a Python program and the necessary files to create a web application that will display the information for all flight departures from Singapore.

The flights should be displayed on the web page labelled **Departure**. It is arranged in ascending order of the departure time, in a table with the following table header:

- Time (that is, departure time)
- To (that is, destination location)
- Flight (that is, flight number)

The resulting web page should be accessed from the “View all Departures” menu option from **Task 4.3**.

Save your Python program as:

TASK_4_5_<your class>_<your name>.py

with any additional files/ subfolders in a folder named:

TASK_4_5_<your class>_<your name>

[2]

Run the web application. Save the departure web page output as:

TASK_4_5_<your class>_<your name>.html

[1]

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Task 4.6

Write a Python program and the necessary files to create a web application that:

- contains a HTML form for user to input `flightNum`
- receives a `flightNum` string from the HTML form, then
- creates and returns a HTML document that enables the web browser to display in a table with heading, the flight details of that `flightNum`.

The flight details are as follows

- `flightNum`
- `departure`
- `destination`
- `departureTime`
- `arrivalTime`

The web page with the HTML form should be accessed from the “Query Flight” menu option from **Task 4.3**.

Save your Python program as:

`TASK_4_6_<your class>_<your name>.py`

with any additional files/ subfolders in a folder named:

`TASK_4_6_<your class>_<your name>`

[4]

Run the web application. **Query for flight SQ111.**

Save the query result web page output as:

`TASK_4_6_<your class>_<your name>.html`

[1]